

## Lab 3 Basic: Timer

### Objective

This lab introduces you to FPGA-based timers using LEDs. You will design a finite-state machine (FSM) in Verilog that controls the LEDs through counting operations. The exercise reinforces FSM design, clock division, and LED sequencing.

### Action Items

#### 1. lab3\_basic.v (100%)

You must design a timer with a finite-state machine (FSM) that controls LED behavior based on the switches.

##### a. I/O list

| I/O       | Connected to | Definition                                                                    |
|-----------|--------------|-------------------------------------------------------------------------------|
| clk       | W5           | Clock signal with the frequency of 100MHz                                     |
| rst       | SW15         | Asynchronous active-high reset signal                                         |
| end_count | SW1          | Ends the <b>PLAY</b> state and triggers transition to the <b>FINISH</b> state |
| led[15:0] | LD15~LD0     | LED status controlled by the FSM                                              |

##### b. FSM Specification

The FSM must include three states: **INITIAL**, **PLAY**, and **FINISH**

- At any time, **rst == 1** resets the FSM to the **INITIAL** state.
- When **rst == 0**, the FSM resumes normal operation.

##### ✓ INITIAL state

- Entered when **rst==1**.
- On the next cycle when **rst==0**, the FSM transitions to the **PLAY** state.
- All LEDs (**LD15~LD0**) are OFF:

LD15 oooooooooooooooooo LD0

(● : LED on, o : LED off)

##### ✓ PLAY state

- If **end\_count** is set to **1**, the FSM transitions to the **FINISH** state synchronously on the next cycle.
- LED behavior:
  - The LEDs are divided into two parts: Upper LEDs (LD15~LD10) and Lower LEDs (LD9~LD0), which operate separately and simultaneously.

- Lower LEDs (LD9~LD0): Timer
  - ◆ Represent a 5-second timer, updating every 0.5 seconds.
  - ◆ Each step lights one additional LED.
  - ◆ When all LEDs are ON, they return to the initial condition and start over.

LD9 ○○○○○○○○○● LD0  
 LD9 ○○○○○○○○●● LD0  
 LD9 ○○○○○○○●●● LD0  
 ...  
 LD9 ●●●●●●●●●● LD0  
 LD9 ○○○○○○○○○● LD0  
 ...  
 (● : LED on, ○ : LED off)

- Upper LEDs (LD15~LD10): Counter
  - ◆ Record how many 5-second cycles have elapsed.
  - ◆ Every 5 seconds, one additional LED in LD15~LD10 lights up.
  - ◆ After LD15 lights up, the FSM transitions to **FINISH** state. For LD15, it lights up for only a moment. You might not see it. Please refer to the demo video.

LD15 ○○○○○○ LD10  
 LD15 ○○○○○● LD10  
 LD15 ○○○○●● LD10  
 ...  
 LD15 ●●●●●● LD10  
 (● : LED on, ○ : LED off)

**Note:** The LED timing rules above must be followed exactly. Failure to do so will result in zero credit for this section.

✓ **FINISH** state

- On entry, all 16 LEDs (**LD15~LD0**) flash together at a rate of **100MHZ/2<sup>27</sup>**.
- LEDs toggle three full cycles (off → on → off → on → off → on).
- After flashing, the FSM returns to the **INITIAL** state.

c. **Demo video:**

Please refer to the attached MP4 file.

**d. Grading Breakdown**

| Functionality          | Weight (100%) |
|------------------------|---------------|
| Reset (rst)            | 5%            |
| INITIAL state          | 15%           |
| PLAY state (LD9~LD0)   | 30%           |
| PLAY state (LD15~LD10) | 30%           |
| PLAY state (end_count) | 15%           |
| FINISH state           | 5%            |

e. You must use the following template for your design:

```
module lab3_basic (  
    input wire clk,  
    input wire rst,  
    input wire end_count,  
    output reg [15:0] led  
);  
    /* Note that output port can be either reg or wire.  
    * It depends on how you design your module. */  
  
    // add your design here  
endmodule
```

**Attention**

- ✓ If you create several modules, merge them into a single Verilog file.
- ✓ DO NOT copy-and-paste code segments from PDFs (hidden characters may cause hard-to-debug syntax errors).
- ✓ Submit **only one** Verilog file, **lab3\_basic.v**.
- ✓ DO NOT submit ZIP files, which will be considered as incorrect format.